Supplementary Material for "A Pilot Study for Text-Column Detection and Reading-Order Evaluation in Traditional Mongolian Historical Archives"

Author One[1] and Author Two[2*]

[1]Department, University, City, 000000, State, Country.
[2*]Department, University, City, 000000, State, Country.

*Corresponding author(s). E-mail(s): author.two@anonymized.edu;
Contributing authors: author.one@anonymized.edu;

This supplementary material provides diagnostic experiments, extended inter-annotator agreement per-page details, and transfer-detector baselines that supplement the main JKSU CIS manuscript.

1 Inter-Annotator Agreement — Per-Page Breakdown

Table 1 reports per-page agreement scores. The per-page breakdown reveals that Ignore-region agreement varies substantially by page (0.000–0.571), consistent with the observation that ambiguous artifact boundaries are the primary source of annotation disagreement. Text-column agreement is consistently high on pages with clear column boundaries (F1 ≥ 0.857 for three of five pages), while the lowest box F1 (0.606 on page 80-48-73-1) reflects a page with irregular short columns and fragments that the two annotators interpreted differently.

2 Supplementary Transfer Baselines

Both RT-DETR-L and DocLayout-YOLO were fine-tuned for 5 epochs under the same train/validation/test split used for YOLOv8n. Because their training budgets are shorter than YOLOv8n (50 epochs) and their pretraining domains are dominated by

Table 1 Per-page inter-annotator agreement metrics on the 5-page independent subset.

Page ID	GT cols	Box F1@0.5	Mean IoU	RO Agree	Ignore F1
80-48-70-2	22	1.000	0.836	0.909	0.000
80-48-62-1(1)	12	0.857	0.766	0.000	0.200
80-48-69-4	2	0.667	0.669	—	0.571
80-48-72-2	10	0.952	0.801	0.889	0.000
80-48-73-1	20	0.606	0.765	0.911	0.400
Aggregate	66	0.852	0.800	0.677	0.286

Table 2 Transfer-detector baselines on the original 11-page test split under the same validation-selected threshold protocol. These diagnostic results suggest that off-the-shelf modern document-layout priors transfer poorly to sparse vertical handwritten archive columns without stronger domain adaptation.

Method	Prec.	Rec.	F1	Mean IoU	RO Acc.
RT-DETR-L + rotation-aware order	0.032	0.270	0.058	0.632	0.847
DocLayout-YOLO + rotation-aware order	0.046	0.031	0.037	0.572	0.000

Table 3 Strict-localization comparison on the original 11-page test split at IoU@0.75. The stricter threshold penalizes boundary errors more heavily.

Method	Prec.	Rec.	F1	Mean IoU	RO Acc.
Heuristic column extraction	0.322	0.447	0.375	0.901	1.000
YOLOv8n + rotation-aware order	0.579	0.434	0.496	0.826	1.000
Faster R-CNN MobileNet + rotation-aware order	0.352	0.289	0.318	0.828	1.000

modern printed document layouts, readers should interpret these results as diagnostic evidence of non-trivial domain transfer rather than as definitive model rankings. A fair-budget retraining comparison is left to future work.

3 Strict-Localization Comparison

4 Inference-Size Sensitivity

Table 4 reports inference-only sensitivity without retraining. Higher inference resolution improves recall at the cost of slower throughput. Note that these values are obtained by running the same 960-trained checkpoint at different test resolutions; full retraining at each image size is recommended before drawing training-configuration conclusions and is left to future work.

Table 4 YOLOv8n inference-only image-size sensitivity on the original 11-page test split. The model is trained at image size 960; inference is run at the listed sizes without retraining.

Inference size	Prec.	Rec.	F1	Mean IoU	RO Acc.
640	0.898	0.579	0.704	0.830	0.900
960	0.956	0.717	0.820	0.770	0.900
1280	0.965	0.737	0.836	0.753	0.900

Table 5 YOLOv8n retrained from scratch at three image sizes (50 epochs each) on the 43-page training set, evaluated on the expanded 54-page test set. Thresholds selected by validation F1.

Training image size	Prec.	Rec.	F1	Mean IoU	RO Acc.
640	0.898	0.852	0.875	0.768	0.997
960	0.902	0.875	0.888	0.770	0.997
1280	0.931	0.813	0.868	0.767	0.997

Table 6 YOLOv8n project-level test performance on the original 11-page test split with different numbers of annotated training pages.

Training pages	Test Prec.	Test Rec.	Test F1	Mean IoU	RO Acc.
7	0.843	0.388	0.532	0.719	1.000
20	0.875	0.691	0.772	0.748	0.900
43	0.956	0.717	0.820	0.770	0.900

5 Multi-Scale Training Ablation

Table 8 reports the effect of retraining YOLOv8n from scratch at three different image sizes with the same 50-epoch budget. Image size 960 achieves the best F1 (0.888), confirming the parameter choice used in the main manuscript. Image size 640 achieves identical F1 (0.875) with a lower threshold (0.20 vs. 0.15 for 960), while 1280 achieves the highest precision (0.931) but lower recall (0.813), consistent with the observation that higher-resolution training encourages more conservative detection. Unlike the inference-only sensitivity table (Table 4), these models were fully retrained at each image size, providing a more reliable picture of the resolution–performance trade-off.

Table 7 Diagnostic column-aware YOLO post-processing on the original 11-page test split.

Setting	Split	Prec.	Rec.	F1	Mean IoU	RO Acc.
Main YOLO threshold 0.35	test	0.956	0.717	0.820	0.770	0.900
Recall-oriented threshold 0.30	test	0.937	0.776	0.849	0.761	0.818
Column-aware prior filter	val	0.929	0.791	0.854	0.779	0.875
Column-aware prior filter	test	0.949	0.730	0.825	0.760	0.818

Table 8 Ablation of ignore-region filtering and rotation-aware reading order at score threshold 0.35 on the original 11-page test split.

Configuration	Prec.	Rec.	F1	RO Acc.	FP/FN	Removed
Full: ignore + rotation-aware order	0.956	0.717	0.820	0.900	5 / 43	0
w/o rotation-aware order	0.956	0.717	0.820	1.000	5 / 43	0
w/o ignore filtering	0.956	0.717	0.820	0.900	5 / 43	0
w/o both	0.956	0.717	0.820	1.000	5 / 43	0

Table 9 Local CPU runtime on the original 11-page test split.

Method	Device	Mean s/page	Median s/page
Heuristic column extraction	CPU	1.843	1.729
YOLOv8n + rotation-aware order	CPU	0.193	0.196
Faster R-CNN MobileNet + rotation-aware order	CPU	0.168	0.162

6 Training-Set Size Scaling

7 Column-Aware Post-Processing Diagnostic

8 Ablation of Ignore-Region Filtering and Rotation-Aware Ordering

9 Runtime Comparison